

THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DARES SALAAM REGION

FORM SIX MOCK EXAMINATION

CHEMISTRY 1

GEPAM science hub

WhatsApp 0748978398

Cod

Time: 3:00 Hours

Monday 3rd February, 2025 P.M

INSTRUCTIONS

1. This paper consists of section A and B with a total of ten (10) questions.
2. Answer **all** questions in section A and two (2) questions from section B.
3. Each question carries **ten (10)** marks in section A and **fifteen (15)** marks in section B.
4. Mathematical tables and non-programmable calculators may be used.
5. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Examination Number** on every page of your answer sheet(s)
7. For calculation you may use the following;
 - $GMV=22.4dm^3$
 - Velocity of light= $3.0 \times 10^8 m/s$
 - Planck's constant= $6.63 \times 10^{-34} Js$
 - Avogadro's number, $A=6.02 \times 10^{23}$ molecules
 - Gas constant, $R= 8.31 Jmol^{-1}K^{-1}$ or $0.0821 atmLmol^{-1}K^{-1}$
 - Absolute temperature= $273K$
 - Atomic masses: $H=1, N=14, O=16, Cl=35.5, Na=23, Ca=40$
 - Density of water = $1g/cm^3$

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This paper consists of 4 printed pages

SECTION A: (70 Marks)**ANSWER ALL QUESTIONS IN THIS SECTION**

1. (a) Write down the electronic configurations of each of the following species by using boxing method
 - (i) Mn^{3+}
 - (ii) Cr
 - (iii) O^{2-}

(3 marks)
- (b) Calculate the wavelength of hydrogen atom of mass $1.674 \times 10^{-27} \text{kg}$ moving at 700cm/s .
(03 marks)
- (c) Describe the Rutherford's observation that;
 - (i) The nucleus has a relatively small diameter compared with that of an atom.
 - (ii) Most of the mass of an atom is concentrated in the nucleus. **(04 marks)**
2. (a) Although the valence shell Electron Pair Repulsion theory (VSEPR) predicts correctly the CH_4 and NH_3 molecular geometries (or shapes), it does not account for the differences in $(\text{H}-\text{C}-\text{H})$ and $(\text{H}-\text{N}-\text{H})$ bond angle. **(03 marks)**
- b) Briefly explain these observations;
 - i) The hydride of fluorine form hydrogen bond but hydrides of chlorine and bromine do not.
 - ii) When molecules of hydrogen chloride are placed together, they do not show induced dipole interactions but they do so when placed with molecules of argon. **(3 marks)**
- c) i) differentiate sp^2 and sp hybridization
 ii) which bond is stronger in the following pairs? Give reason
 $\text{C} \equiv \text{O}$ or $\text{C}=\text{O}$; $\text{C}-\text{N}$ or $\text{C}-\text{O}$; $\text{C}-\text{C}$ or CC **(04 marks)**
3. (a) (i) Use kinetic molecular theory to explain the origin of gas pressure.
 (ii) Can the speed of a given molecule in a gas double at constant temperature? Explain your answer **(03 marks)**
- (b) A sample of tear gas is released in the last row seats in a form five class. At the same time a sample of laughing gas is released in the first row seats. After a time the last 10 rows of students were crying and the first 20 rows of students were laughing. If the laughing gas is N_2O with molar mass of 44, what is the molar mass of the tear gas? **(04 marks)**
- (c) A volatile organic compound weighs 0.2g. On heating in the Victor Meyer's tube it displaces 30mls of air at 27°C and 756mmHg atmospheric pressure. Determine the molecular mass of the compound given that the aqueous tension at 27°C is equal to 26mmHg. **(3 marks)**
4. a) i) State Hess's law of constant heat summation.
 ii) Describe two applications of Hess's law. **(02 marks)**
- b) Calculate the ΔH for the reaction $\text{C}_2\text{H}_{4(g)} + \text{H}_{2(g)} \rightarrow \text{C}_2\text{H}_{6(g)}$ from the data below;

$$\text{C}_2\text{H}_{4(g)} + 3\text{O}_{2(g)} \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O}_{(l)} \quad \Delta H = -1411 \text{KJmol}^{-1}$$

$$\text{C}_2\text{H}_{6(g)} + \frac{7}{2}\text{O}_{2(g)} \rightarrow 2\text{CO}_{2(g)} + 3\text{H}_2\text{O}_{(l)} \quad \Delta H = -1560 \text{KJmol}^{-1}$$

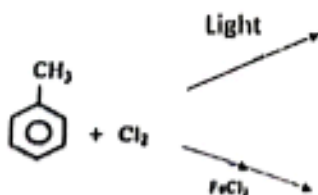
$$\text{H}_2 + \frac{1}{2}\text{O}_{2(g)} \rightarrow \text{H}_2\text{O}_{(l)} \quad \Delta H = -286 \text{KJmol}^{-1} \quad \text{(03 marks)}$$

- c) A 1.150g of sucrose was burnt in a bomb calorimeter. If the temperature rose from 23.42°C to 27.64 °C and the heat capacity of the calorimeter was 4.90Kj °C⁻¹. What was the heat of combustion of sucrose (C₁₂H₂₂O₁₁) in kilojoules per mole? **(04 marks)**
5. (a) Nitrogen and oxygen gases combine reversibly at high temperature with absorption of heat to form nitrogen monoxide gas.
- Write a balanced chemical equation for the reaction
 - Write the expression for the equilibrium constant K_p for the reaction. **(2 marks)**
- (b) When solid silver is added to a solution containing Ag⁺, Fe²⁺ and Fe³⁺ with initial concentration 0.4M, 0.2M and 0.6M respectively; the following reversible reaction occurs.
- $$Ag_{(aq)}^+ + Fe_{(aq)}^{2+} \rightleftharpoons Fe_{(aq)}^{3+} + Ag_{(s)}$$
- If the K_c for this reaction is 5.96 at 25°C, calculate the equilibrium concentration of each ion in the equilibrium solution. **(04 marks)**
- (c) The following equilibrium is set up in solution by dissolving cobalt (II) chloride crystals in water to form pink species [Co(H₂O)₆]²⁺ and then adding concentrated hydrochloric acid until the solution becomes blue.
- $$[Co(H_2O)_6]_{(aq)}^{2+} + 4HCl_{(aq)} \rightleftharpoons [CoCl_4]_{(aq)} + 6H_2O_{(l)}$$
- Does the reaction cease when the solution become blue? Explain.
 - The forward reaction is endothermic, state and explains colour change observed by cooling the mixture.
 - Mention one way of reversing the change caused by cooling the reaction **(04 marks)**
6. (a) A form six student from Mtakuja Secondary School dropped pieces of sodium metal in water and observed the interested phenomena that trigged him to heat another sodium metal in excess air that is free of carbon dioxide and observed another interesting phenomenon that made his day. Explain the observation made by a student in both phenomena. **(04 marks)**
- b) Explain these facts with suitable equations where necessary
- The aqueous solution of iron (III) chloride is acidic.
 - Gaseous iron (III) chloride at low temperature exists as a dimer and it is a covalent compound.
 - Iron (III) carbonate does not exist
 - Red hot iron decomposes steam reversibly. **(06 marks)**
7. (a) A 1% solution of sodium chloride freezes at -0.604°C. Calculate the degree of dissociation of the sodium chloride if the molal freezing point depression constant of water is 1.86°C/m. **(3 marks)**
- b) Quinine was first drug in malaria treatment and it remains the treatment of choice for severe cases. A solution prepared by mixing 10g of quinine in 50ml of ethanol has a freezing point of 1.55°C below that of pure ethanol. Determine the molar mass of quinine, given that the density and cryoscopic constant for ethanol are 0.789g/ml and 1.99 °C/m respectively. What assumption(s) have you made in your calculation. **(04 marks)**
- c) What will be the freezing point and boiling point of an aqueous solution containing 55.0g of glycerol, C₃H₅(OH)₃, and 250g of water? Given K_b(H₂O) = 0.51°C/m and K_f(H₂O) = 1.86 °C/m **(03marks)**

SECTION B (30 marks)

ANSWER ANY TWO QUESTIONS FROM THIS SECTION

8. a) Mention four (4) advantages of chemical fertilizers over manure **(04 marks)**
b) A farm requires 120kg of nitrogen. What mass of urea fertilizer labelled 37-0-0 needs to be applied to the soil to meet this demand? Show clearly your calculations **(05 marks)**
c) You are a specialist who wants to determine the quality of lines so that you can advise farmers and other stake holders on the best liming materials. Show how you can go about on the calculation of the neutralizing value of Calcium Hydroxide $(\text{Ca}(\text{OH})_2)$. **(06 marks)**
9. (a) Explain briefly the following concept:
i. Concentrated sulphuric acid is necessary for nitration of benzene.
ii. Brominating benzene takes place in the presence of Lewis acid while that of hydroxybenzene does not require presence of Lewis Acid.
iii. Despite chlorine being an electron withdrawing group, it directs an incoming group to the ortho and para positions in electrophilic aromatic substitution reactions. **(03 marks)**
- (b) Give one chemical test to distinguish given sets of reagents. Support your answer with a chemical equation.
i. Benzene and ethene
ii. 3-chloro-2-methylpent-2-ene and 1-chloropropane.
iii. Bromobenzene and bromomethane. **(06 marks).**
- (c) Two isomeric hydrocarbons K and L have the molecular formula C_9H_{12} . On oxidation K gives a monocarboxylic acid which when heated with excess soda lime yields benzene. When L is oxidized, it gives tricarboxylic acid which can undergo nitration to give a mono-derivative. What are the structure formula of K and L? Write all the chemical reactions occurred **(06 marks).**
10. (a) A form six students gave the following names for various substituted aliphatic hydrocarbons;
(i) 2-methyl-3-bromobutane,
(ii) 3,3-dimethyl-2-chlorobutane,
(iii) 4-chloro-3-bromo-2-pentene,
(iv) 2-methyl-4-butyne.
- The names indicate the formulae of the substituted aliphatic hydrocarbons but do not strictly obey IUPAC rules. Draw the structure suggested by the incorrect names and assign the correct name for each compound. **(04marks)**
- (b) Using benzene as a starting material, show how you can prepare each of the following.
i. Tribromobenzene
ii. Propylbenzene
iii. Benzoic acid **(06 marks)**
- (c) Write the structures and the names of five products in the following reaction

**(05 marks)**

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FORM SIX MOCK EXAMINATION 2025
CHEMISTRY I MARKING SCHEME

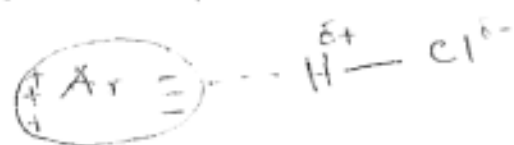
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2(a) So bonds are slightly constricted and form pyramidal shape with bond angle 107°
0/1/2

(b)(i) This is due to highest electronegativity of F which cause HF to be polar and hence H become positively charged and can form hydrogen bond.
0/1/2

(ii) HCl is polar hence the dipole already exist and cannot be induced. But argon is neutral hence the dipole can be induced in Ar when it is close to polar HCl molecule.



0/1/2

(c)

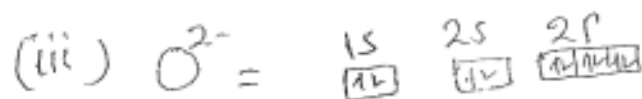
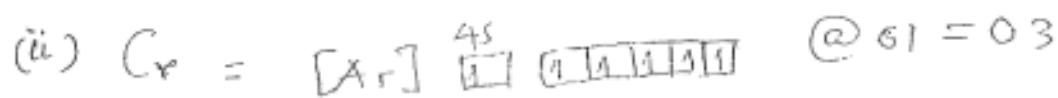
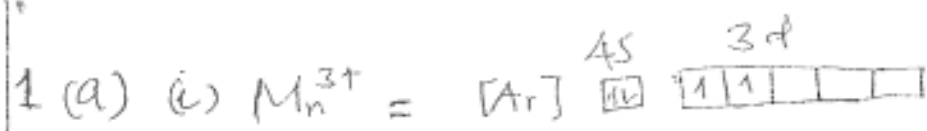
(i)	sp^2 hybridization	sp hybridization
1	Have 33% s-character and 66% p-character	- Have 50% s-character and 50% p-character
2	Involves one s-orbital and two p-orbitals	- Involves one s-orbital and one p-orbital

(ii) $C \equiv O$ is stronger than $C=O$ because triple is stronger than double bond since $\equiv$ involves six shared electrons and $=$ involves 4 shared electrons between nuclei of C and O

$C=O$ is stronger than $C \equiv N$ because oxygen is more electronegative than nitrogen

$C=C$ is stronger than $C-C$ because double bond contain sigma and pi-bond that is formed by sharing 4 electrons between C atoms.

(02)



(b) $\lambda = \frac{h}{mc} \dots \dots \dots 01$

$= \frac{6.63 \times 10^{-34} \text{ Kg m}^2 \text{ s}^{-1}}{1.674 \times 10^{-27} \text{ Kg m s}^{-1} \times 7} \dots \dots 01$

$\lambda = 5.658 \times 10^{-8} \text{ m} \dots \dots \dots 01$

(c) (i) The nucleus is smaller because it only contains the dense, massive particles (protons and neutrons). $\dots \dots \dots 02$

(ii) Because the nucleus contains the protons and neutrons which are more massive than electrons $\dots \dots \dots 02$

2(a) Due to repulsive force variation of the bond pairs and lone pairs distorted from angular geometry and organize themselves in such a way that repulsion will be maximum and stability will be maximum. $\dots \dots \dots 01$

So, in case of CH_4 , there are four bond pairs and no lone pair. It remains in its regular geometry; tetrahedral with bond angle 109.5°

In NH_3 there are three bond electron pairs and one lone pair. Since repulsion between bond pair and lone pair is more than that of two bond pair $\dots \dots \dots 01/2$

3. (i) (a) pressure of the gas is due to collision of gas molecules themselves and with the wall of the container. 0/2

(ii) Yes. Because at any given temperature, molecules move with independent speed. 0/2

(b) From Graham's Law:

$$\frac{R_A}{R_B} = \sqrt{\frac{M_{rB}}{M_{rA}}} \quad 0/1$$

$$\frac{10 \text{ units}}{20 \text{ units}} = \sqrt{\frac{44}{M_r}} \quad 0/1$$

Molar mass of tear gas = 176 g mol^{-1} 0/1

(c) Pressure of organic compound = $P_T - P_w$
 $= (0.995 - 0.04) \text{ atm}$
 $= 0.96 \text{ atm} \quad 0/1$

$$PV = nRT \quad 0/1$$

$$PV = \frac{m}{M_r} RT$$

$$M_r = \frac{mRT}{PV} \quad 0/1$$

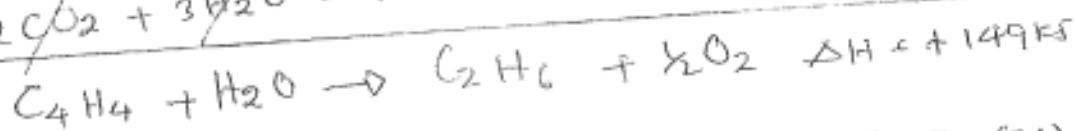
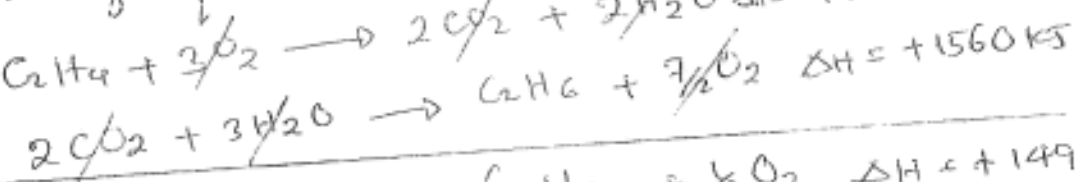
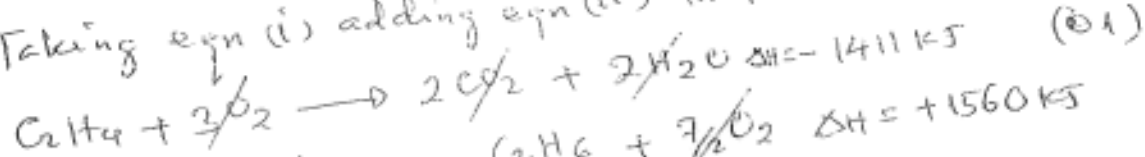
$$= \frac{0.2 \text{ g} \times 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1} \times 300 \text{ K}}{0.96 \text{ atm} \times 0.03 \text{ L}}$$

$$M_r = 170.9 \text{ g/mol.}$$

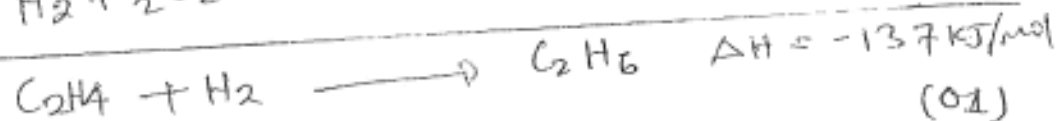
4(a)(i) Hess's law of constant heat summation states that "The heat of reaction is constant regardless the route taken to progress of the reaction"

(ii) To estimate the enthalpies of reactions
 - To determine the overall energy required for a chemical reaction @ 0.1 = 0.2.

(b) Taking eqn (i) adding eqn (ii) in reverse



Add eqn (iii)



(c) Mass of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) = 1.150g

$$T_1 = 23.42^\circ\text{C}$$

$$T_2 = 27.64^\circ\text{C}$$

$$\Delta T = T_2 - T_1 = 27.64 - 23.42 = 4.22^\circ\text{C}$$

Heat capacity of calorimeter = 4.90 kJ/°C

$$\text{From } Q_{\text{calorimeter}} = C \times \Delta T$$

$$= 4.90 \times 4.22 = 20.7 \text{ kJ} \quad - 0.1$$

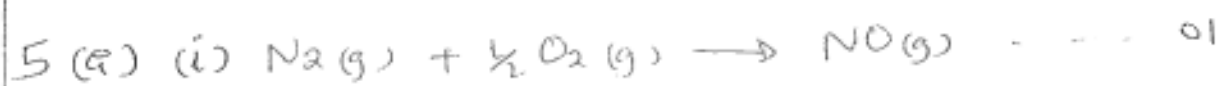
Heat released in the combustion = Heat gained by calorimeter

$$n_{\text{sucrose}} = \frac{m}{M_r} = \frac{1.150 \text{ g}}{342 \text{ g/mol}} = 0.0033 \text{ mol.}$$

$$4 (c) \text{ Heat of combustion} = \frac{-20.7 \text{ kJ}}{0.0033 \text{ mol}} \quad \dots (0.1)$$

$$= -6160 \text{ kJ mol}^{-1}$$

∴ Heat of combustion of sucrose is -6160 kJ/mol (0.1)



(ii) $K_p = \frac{p'_{\text{NO}}}{p'_{\text{N}_2} \times p'_{\text{O}_2}^{1/2}} \quad \dots 001$

(b) $[\text{Ag}^+] = 0.4 \text{ M}$

$[\text{Fe}^{2+}] = 0.2 \text{ M}$

$[\text{Fe}^{3+}] = 0.6 \text{ M}$

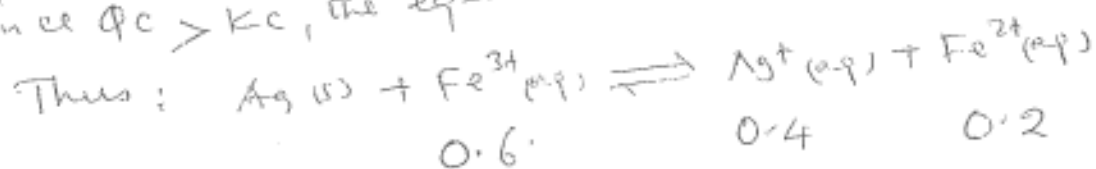
$$Q_c = \frac{[\text{Fe}^{3+}]}{[\text{Ag}^+][\text{Fe}^{2+}]}$$

$$= \frac{0.6 \text{ M}}{0.4 \text{ M} \times 0.2 \text{ M}}$$

$$= 7.5 \text{ M}^{-1}$$

(0.1)

Since $Q_c > K_c$, the equilibrium shifts to the left.



0.6	0.4	0.2
0.6 - x	0.4 + x	0.2 + x

$$K_c = \frac{1}{K_c} = \frac{[\text{Ag}^+][\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} \quad (0.1)$$

$$5(b) \quad \frac{1}{5.96} = \frac{(0.4+x)(0.2+x)}{(0.6-x)} \quad (01)$$

$$X = 0.03M$$

Equilibrium concentration

$$[Ag^+] = 0.2 + 0.03 = 0.23M$$

$$[Fe^{2+}] = 0.4 + 0.03 = 0.43M$$

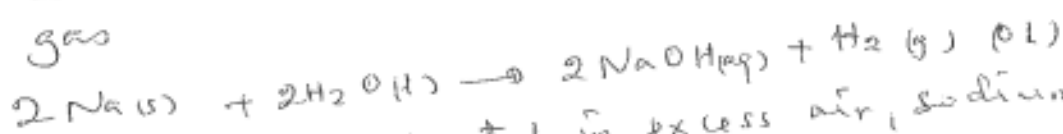
$$[Fe^{3+}] = 0.6 - 0.03 = 0.57M$$

(c) (i) Yes: because there is excess product than reactant at equilibrium.

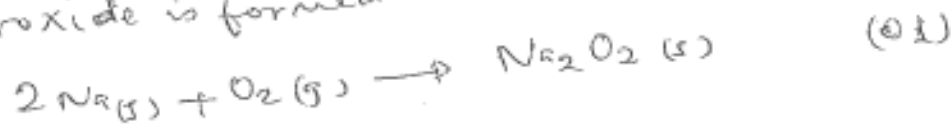
(ii) Colour will change from blue to pink on cooling i.e. more $[Co(H_2O)_6]^{2+}$ will be formed

(iii) By adding conc. HCl. @ 01 = 03

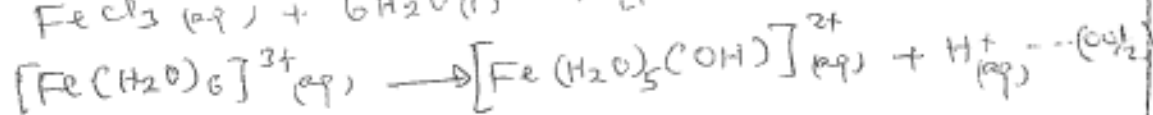
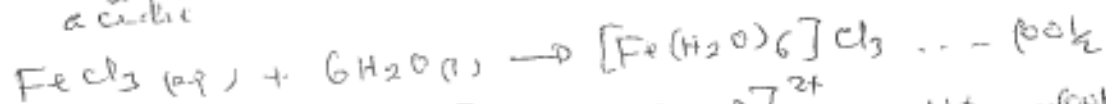
6(a) Sodium metal pieces reacts violently with water to form sodium hydroxide and hydrogen gas (01)



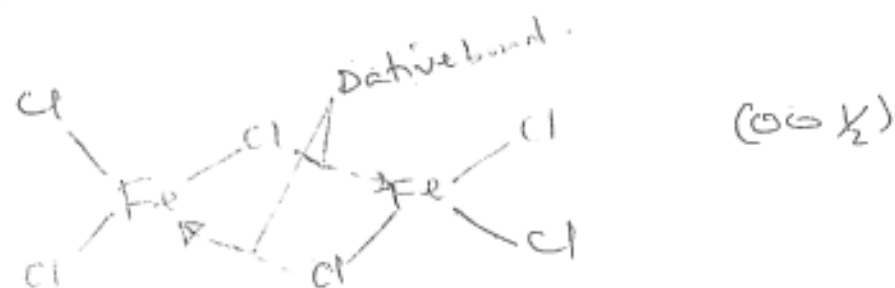
When Sodium is heated in excess air, sodium peroxide is formed. 01



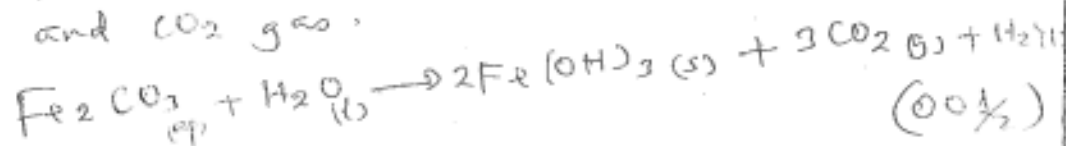
(b) (i) Iron(III) chloride in aqueous solution form - hexaaqua iron(III) ion complex which decompose to give H^+ ions and hence the solution become acidic (01/2)



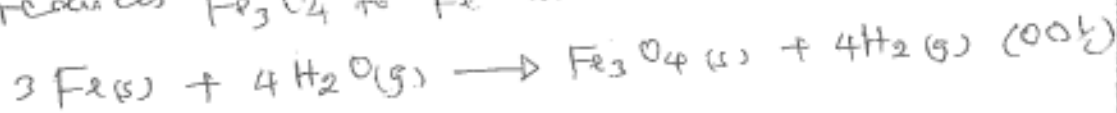
6 (b)(ii) Anhydrous gaseous iron(III) chloride vapourises at low temperature due to formation of dative bond between lone pairs in Cl atoms and the empty d-orbital in Fe. (01)



(iii) Iron(III) carbonate does not exist in a solution because, iron(III) ions bonds to water molecules and exists as $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$. Due to strong interaction between Fe^{3+} ions and aqua ligand, the complex dissociates into negatively charged complex and H^+ ions because CO_3^{2-} is a base and captures H^+ ions to form hydroxyl complex which produces H_2CO_3 and iron(III) hydroxide. The weak carbonic acid decomposes to water and CO_2 gas. 01



(iv) Red hot iron decompose steam to form triiron tetra oxide and hydrogen gas. H_2 at high temperature reduces Fe_3O_4 to Fe and H_2O . (01)



$$7 (a) \Delta T_f = i K_f \cdot m$$

$$\Delta T_f = i \cdot K_f \times \frac{n_s}{m_{\text{solvent}}(\text{kg})} \quad (10 \frac{1}{2})$$

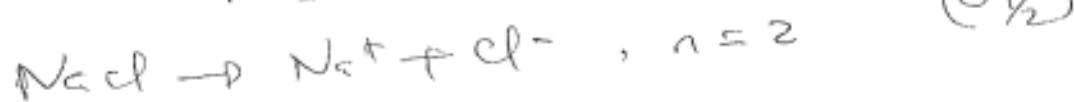
$$n_s = \frac{3g}{58.5 \text{ g mol}^{-1}} = 0.017 \text{ mol}$$

$$\begin{aligned} \Delta T_f &= T_{\text{solvent}} - T_{\text{solution}} \\ &= 0^\circ\text{C} - (-0.604^\circ\text{C}) \\ &= \underline{0.604^\circ\text{C}} \quad (1) \end{aligned}$$

$$0.604 = \frac{i \times 1.86 \times 0.017}{0.0099 \text{ kg}}$$

$$i = 1.89 \quad (1)$$

$$\alpha = \frac{i-1}{n-1}$$



$$\alpha = \frac{1.89-1}{2-1} = 0.89 \text{ or } 89\% \quad (1)$$

$$(b) w_{\text{solute}} = 10g$$

$$w_{\text{solvent}} = 0.789 \text{ g/ml} \times 50 \text{ ml} = 39.45 \text{ g}$$

$$\Delta T_f = 1.55^\circ\text{C}$$

$$K_f = 1.99^\circ\text{C/m}$$

$$M_{\text{solute}} = ?$$

$$\text{From } \Delta T_f = \frac{K_f \times w_{\text{solute}} \times 1000}{M_{\text{solute}} \times w_{\text{solvent}}} \quad (1)$$

$$7(b) \quad 1.55 = \frac{1.99 \times 10 \times 1000}{M_{\text{solute}} \times 39.45}$$

(10/1)

$$M_{\text{solute}} = \frac{1.99 \times 10 \times 1000}{1.55 \times 39.45}$$

$$= 325.4 \text{ g/mol}$$

$\therefore$ Molar mass of quinine is 325.4 g/mol . (01/2)

$$(c) \text{ Molar mass of solute, } (C_3H_5(OH)_3) = (3 \times 12) + 5 + 17 \times 3 = 92 \text{ g/mol}$$

$$\Delta T_f = ?$$

$$\Delta T_f = K_f \times \frac{w_{\text{solute}}}{M_{\text{solute}}} \times \frac{1000}{w_{\text{solvent}}} \quad \dots \dots \dots 00 \frac{1}{2}$$

$$\Delta T_f = 1.86 \times \frac{55}{92} \times \frac{1000}{250} = 4.45^\circ \text{C} \quad \dots \dots \dots 00 \frac{1}{2}$$

$$\Delta T_f = T_f - T_{f \text{ soln}}$$

$$4.45^\circ \text{C} = 0 - T_{f \text{ soln}}$$

$$T_{f \text{ soln}} = -4.45^\circ \text{C} \quad \dots \dots \dots (00 \frac{1}{2})$$

$\therefore$ Freezing point of solution is -4.45°C (00/2)

$$\Delta T_b = K_b \times \frac{w_{\text{solute}}}{M_{\text{solute}}} \times \frac{1000}{w_{\text{solvent}}} \quad \dots \dots \dots 00 \frac{1}{2}$$

$$\Delta T_b = 0.51 \times \frac{55}{92} \times \frac{1000}{250} = 1.22^\circ \text{C} \quad \dots \dots \dots 00 \frac{1}{2}$$

$$\Delta T_b = T_b \text{ soln} - T_b \text{ solv}$$

$$1.22^\circ \text{C} = T_b \text{ soln} - 100^\circ \text{C}$$

$$T_b \text{ soln} = 101.22^\circ \text{C}$$

$\therefore$ Bp of solution is 101.22°C (01)

8(a) - Fertilizers are nutrient specific while manure is not

- Fertilizers are not bulky hence easy to be transported and stored than manure.

- Fertilizers provides nutrients when applied but manure depend on other factors like - microorganisms.

- Fertilizers are pleasant to use compared to manure.

@ 1 = 0.4

(b) percentage of Nitrogen in the fertilizer is 37.

No phosphorus

No potassium

Let the amount in kg of fertilizer needed be X (01)

$$37\text{kg of N} \equiv 100\text{kg of fertilizer}$$

$$120\text{kg} \equiv X \quad (01)$$

$$X = \frac{120\text{kg} \times 100\text{kg}}{37\text{kg}} = 324.32\text{kg} \quad (01)$$

∴ Therefore, 325.32 kg of the fertilizer is needed to be applied to meet the demand of the farm. (02)

(c) ^{molar} Mass of $\text{Ca}(\text{OH})_2 = 74\text{g/mol}$

molar mass of $\text{CaCO}_3 = 100\text{g/mol}$

Grams of CaCO_3 equivalent to 100g of a given -
liming material = X

Since 1 mole of $\text{Ca}(\text{OH})_2$ is equivalent to 1 mole of CaCO_3 then,

$$74\text{g of } \text{Ca}(\text{OH})_2 \equiv 100\text{g } \text{CaCO}_3$$

$$100\text{g of } \text{Ca}(\text{OH})_2 \equiv X\text{g of } \text{CaCO}_3$$

$$8(c) \quad x = \frac{100g \times 100g}{74g} \quad \text{--- (01)}$$

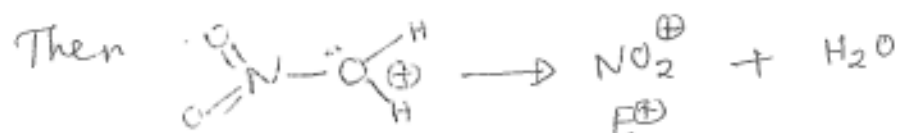
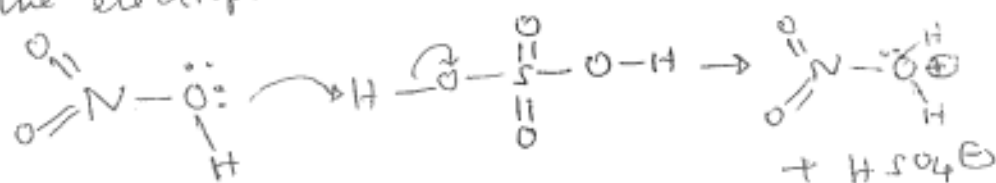
$$x = 135g \quad \text{(01)}$$

$$NV = \frac{\text{grams of pure } CaCO_3 \text{ equivalent to } 100g \text{ of a given lining material}}{100g \text{ of } CaCO_3} \times 100\% \quad \text{(01)}$$

$$NV = \frac{135g}{100g} \times 100\% = 135\% \quad \text{--- (01)}$$

∴ Therefore the neutralizing value of $Ca(OH)_2$ is 135% (02)

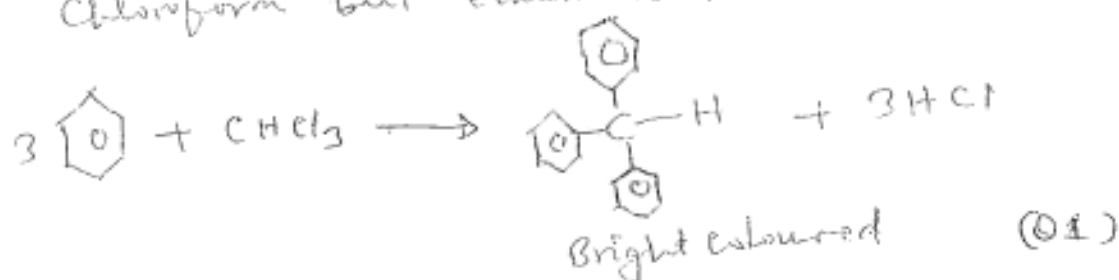
9(a)(i) Sulphuric acid acts as a catalyst in the nitration of benzene. Sulphuric acid in the mechanism is used to form the electrophile (01)



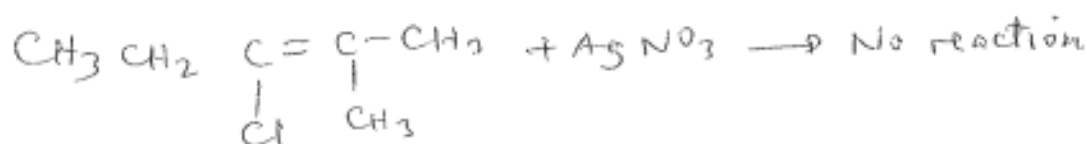
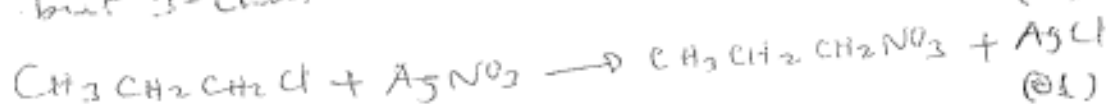
(ii) Hydroxyl group donate electrons to benzene by +M effect hence increases reactivity of benzene. That is why hydroxybenzene bromination don't require the presence of Lewis acid. (01)

9(a)(iii) Chlorine is a deactivator because it withdraws electrons by $-I$ effect but also it donates electrons by $+M$ effect due to presence of lone pair. Due to $+M$ effect, chlorine becomes ortho or para directing. - - - (01)

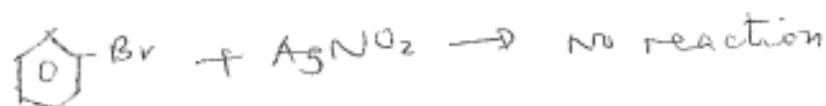
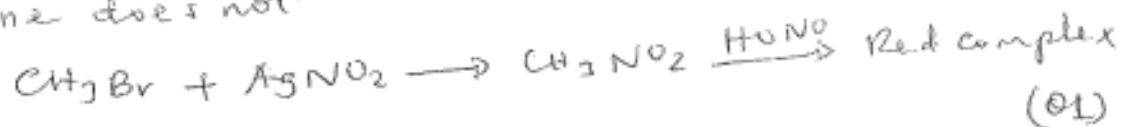
(b) (i) Benzene forms bright coloured complex with chloroform but ethane does not.



(ii) 1-chloropropane forms white ppt with AgNO_3 but 3-chloro-2-methylpent-2-ene does not. (01)



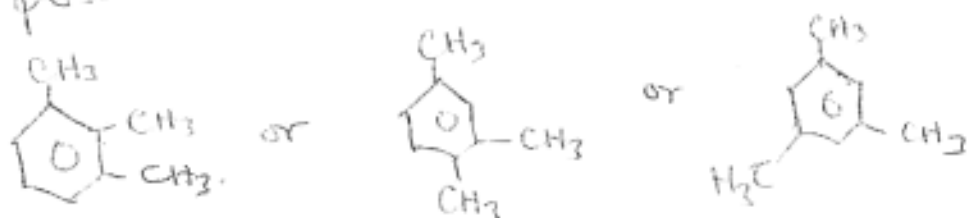
(iii) Bromomethane reacts with AgNO_2 followed by nitrous acid to give red complex but bromobenzene does not.



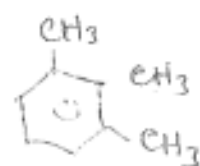
(c) K on oxidation gave a monocarboxylic acid; hence it is alkyl benzene. K is $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_3$

Q(c) L on oxidation gave tricarboxylic acid,
hence it contain three alkyl groups att
... Ched on benzene.

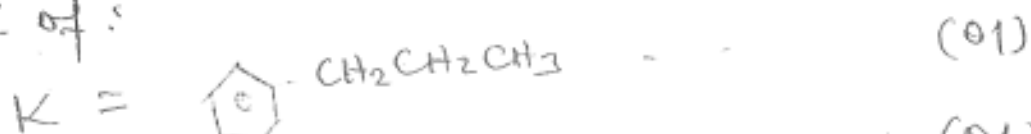
The possible structures of L are:



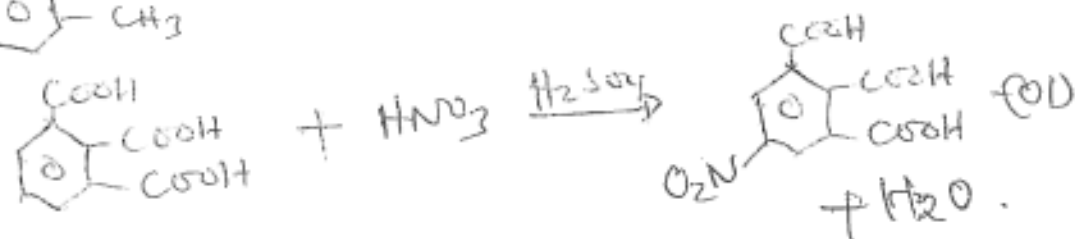
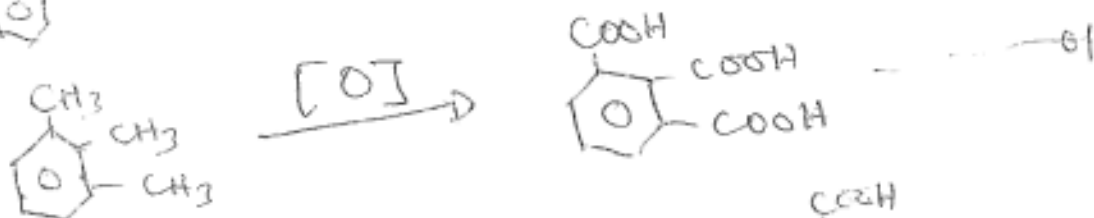
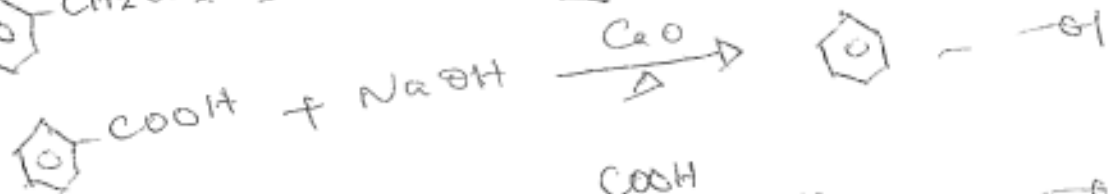
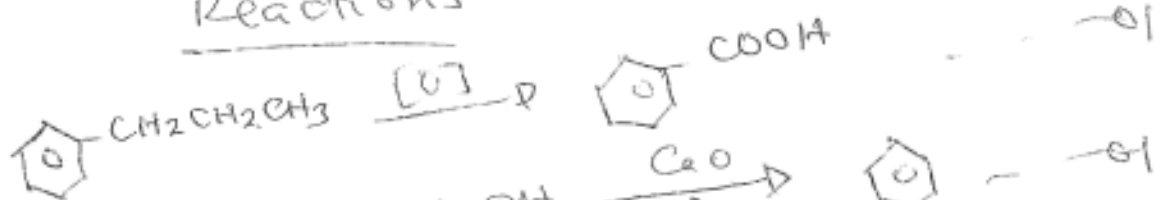
The correct s.f of L is

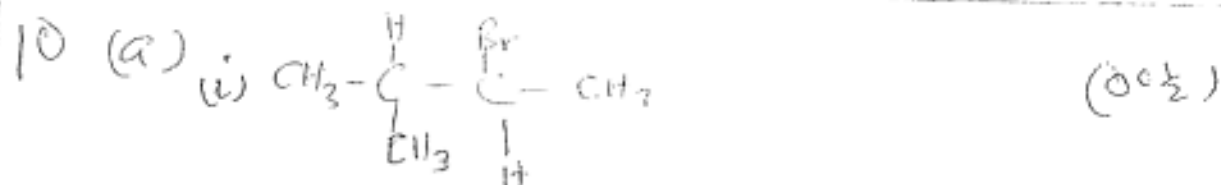


S.f of:



Reactions

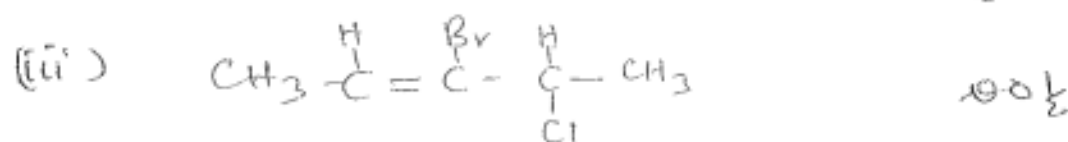




2-Bromo-3-methylbutane (0.5)



2-chloro-3-dimethylbutane (0.5)



3-Bromo-4-chloropent-2-ene 0.5



3-methylbut-1-yne (0.5)

